World Bank Indicators on electricity access and economic development 2000-2023

Sample Report by Manali D



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Executive summary

Access to electricity is crucial for socio-economic development. Such access also allows populations to seek education, use technology, improve their financial condition and also improve health with an improvement in their overall living standards. It is therefore important to understand the role electricity plays not just in developed countries but also developing countries for a span of years to understand any correlational and causal links between access to electricity and economic and social development. Even though cultural and political context is also important as it contributes to the development of a nation, this report uses associations with access to electricity and its associations with health, digital connectivity, education and population characteristics. Descriptive statistics, correlation analysis, country-level comparisons and panel data for regression models is used to investigate patterns in the obtained data.

Data is assembled using APIs from publicly available datasets on the World Bank Website. The cleaned dataset consisted of 6,916 observations across 18 columns with 266 countries represented in the data. A time period of 23 years is reported under this data, from 2000 to 2023 for the indicators under consideration. 6,786 observations from country level data are with country codes. Remaining observations that represent five

regional or income aggregates across 23 years plus are excluded from the country level data.

Key findings of this report include-

1. Access to electricity is positively correlated to GDP per capita. Pooled country- year Pearson correlation was found to be 0.384, indicating a moderate positive linear association. Spearman correlation for rank based associations are 0.847,higher than mentioned 0.384
2. Electricity access is positively associated with broader development outcomes. Pearson's correlation for electricity access and life expectancy is 0.829 and between electricity access and infant mortality is -0.827. These associations reveal a deeper patterns between electricity access and better health outcomes.
3. Access to electricity is closely related to access to basic drinking water. The Pearson correlation between electricity access and basic drinking water access is 0.909 and drinking water access has a strong negative correlation with infant mortality at -0.846.
4. Electricity access and drinking water outcomes show varying relationships across regions. Sub-Saharan Africa has lowest regional mean of access to electricity in the regional sample data, stated at 40.06% , Europe , Central Asia & North America have near-universal average access. Detailed variations across different geographical areas were not clearly established.

5. Access to electricity had a positive correlation with development. In a two way fixed effects model, a percentage point increase in electricity access is associated with approximately 0.65% higher GDP per capita after accounting foe country and yearly fixed effects. The pooled OLS for this is higher.

6. Fixed-effects estimates are smaller compared to poole effect estimates. If country specific characteristics are considered and common year effect is accounted for, the associations also then differ after this consideration is made.

Introduction

Burke et al., 2018 state in their study that electricity access and gross domestic product or GDP are strongly correlated. Even though literature successfully captures associations such as correlation there is limited evidence on the causal links between electricity access and GDP of a country. Tong et al., 2019 noted that electricity access not only contributes to a nation's GDP but also shows positive correlations between education ,improved financial conditions, industrialization and the infrastructure of a nation. This document will record explored relationships between electricity and economic development, using GDP per capita as the primary economic indicator.

Significance

Research Question: What is the association between access to electricity and GDP per capita across 266 countries between 2000-2023 and further finding their association with other developmental indicators ?

Sub questions addressed-

1.How does access to electricity vary across countries between 2000 -2023?

2.How strongly is electricity access associated with GDP per capita?

3How does access to electricity relate to health, education, infrastructure, and digital connectivity?
4.How do the above stated relationships differ across regions defined by the World Bank and whether those differences remain after accounting for country and year effects?

Objectives:
Five principal objectives for this report are as follows-

- Using selected World Bank development indicators create country-year panel data
- Reporting the distribution, coverage, and missingness for the selected indicators.
- Examining relationships between access to electricity and other development measures. Use Pearson and Spearman correlation tests.
- Comparing development patterns across geographical regions.
- Estimating pooled OLS and two-way fixed-effects panel regression models to examine whether selected associations persist after accounting for country-specific and common year effects.

Study and its scope

This analysis utilizes indicators available from World Bank data between 2000–2023. The cleaned dataset includes 15 indicators as well as country identifiers and year information.The empirical analysis focuses on country-level observations. Regional and income aggregates are retained in the full dataset for completeness but excluded from country-level statistical analysis.

The is an observational type of study and identifies relationships between independent and dependent variables of interest. Further we comment on the patterns noted; it does not establish causality in electricity access directly with economic or social outcomes.

Data pipeline, acquisition and analysis

Data pipeline was created and data quality checks were performed:

1. Data Retrieval for selected World Bank indicators.
2. Cleaned and standardized country as well as the indicator records.
3. Kept missing data labeled under 'NaN'.

4. Merged data indicators using country identifiers and year.
5. Created country-year panel.
6. Keep country-level records from regional and income aggregates separate for appropriate analyses.
7. Performed descriptive statistics, correlation matrices, regression outputs, and visualizations.

Assessing and handling missing data:

Dataset Overview

The total number of observations in the dataset are across 6,916 rows and 18 columns, with a total of 266 entities and 15 indicators. 6,786 observations represent country-level data, while 130 do not. In other words, these 130 observations correspond to five aggregate entities. The dataset spans 2000–2023 for the stated period for the report, although individual analyses do use different available-year ranges. Refer Table 1.

Missing values

Missing data are not tackled but retained, refer Table 2. Analyses used only available observations for each indicator or indicator pair, so sample sizes varied across analysis and models. Missing ness was reported through descriptive summary analysis. Therefore, results should be interpreted with consideration of such inconsistencies in the data availability.

Descriptive statistics : Univariable analysis was performed for selected indicators, refer Table 4.

The mean GDP per capital was found to be 14,968.643. Population mean was reported at 266,106,826.895. 80% of the population was reported having secondary school enrollment. 80% among 6191 sample population has access to electricity. 39% of 5509 were found to make use of the internet. 79.5% of 6118 sample population had cellular subscription, 86% of the sample population of 6298 had access to basic drinking water. 51.9% of the people out of the 6525 total sample composed of the urban population. 30% of 5626 were found to utilize renewable energy. 70.46% of 6525 had reported life expectancy at birth. 27.7% reported infant mortality per 1000 live births of the total sample of 6000 people. Smoking prevalence was 38.6 and 12.0 among 1656 men and 1656 women, respectively. Alcohol consumption was 5.3 per capita for 4815 observations. Diabetes had the majority percentage of missing data in the data set with only 517 observations , prevalence for which was reported at 92.41.

Bivariable Analysis : Bivariable Analysis was performed for selected indicators, refer Table 5.

Under this relationships between indicators were used to find association using scatterplots , Pearson and Spearman correlation coefficients. Strong positive correlations were observed between electricity access and life expectancy at $r = 0.830$ and $\rho = -0.872$, between electricity access and access to drinking-water at $r = 0.909$, $\rho = 0.884$. Where negative associations were found between electricity access and infant mortality at $r = -0.827$, $\rho = -0.872$ and infant mortality and drinking water access at $r = -0.846$ and $\rho = -0.883$. Revealing development-related patterns for the mentioned indicators.

Multivariable analysis: Multivariable analysis was performed using Pooled Ordinary least squares (OLS) and 2-way mixed effects analysis, refer Table 6

Pooled Ordinary least squares (OLS) equation-

$$Y(kt) = \beta 0 + \beta 1 X(kt) + \varepsilon(kt)$$

Two way mixed effects analysis equation-

$$Y(kt) = \beta 1 X(kt) + \boldsymbol{\alpha(k)} + \gamma(t) + \varepsilon(kt)$$

Y (kt) = **Outcome of interest for country k in year t also known as dependent variable of interest**

X(kt) = **Independent variable of interest**

B1 = **Coefficient**
α(k) = Country fixed effects
γ(t) = **Years fixed effects**
ε(kt) = **Error term**

Ordinary least squares (OLS) for pooled data and two-way fixed-effects models were used for performing regression analysis to examine associations between mentioned development indicators. The fixed-effects models were explored for country-specific and year-specific effects, for which standard errors were clustered for countries. These estimates helped compare and distinguish overall pooled associations from relationships based on within-country changes over time.

Results

1. Electricity access and log GDP per capita: A one-percentage-point increase in population with electricity access was associated with approximately 4.00% higher GDP per capita in the pooled OLS model.

2. Electricity access and life expectancy: A one-percentage-point increase in population with electricity access was associated with 0.254 additional years of life expectancy in the pooled OLS model.

3. Electricity access and infant mortality: A one-percentage-point increase in population with electricity access was associated with 0.711 fewer infant deaths per 1,000 live births in the pooled OLS model.

4. Basic drinking-water access and infant mortality: A one-percentage-point in population with increase in basic drinking-water access was associated with 1.192 fewer infant deaths per 1,000 live births in the pooled OLS model.

5. Basic drinking-water access and life expectancy: A one-percentage-point increase in population with basic drinking-water access was associated with 0.413 additional years of life expectancy in the pooled OLS model.

6. Log GDP per capita and secondary school enrollment: A one-unit increase in log GDP per capita was associated with a 15.19-percentage-point increase in secondary school enrollment in the pooled OLS model.

Discussions

The findings suggest that access to electricity is closely associated with development outcomes, including health, where the economy of the country stands at internet use and digital connectivity. Stronger correlations indicate that essential infrastructure and development indicators usually go

hand-in-hand across country-year observations. While, smaller fixed-effects coefficients suggest that pooled associations partly reflect persisting differences between nations and those shared differences change over time. Country and year fixed effects account for time-invariant characteristics of countries and common year-specific influences; this however does not eliminate underlying confounding. Causal links are weak. Variation in sample sizes, missing data, indicator coverage, and current-US dollar GDP measurements and their consistency with with the definitions over time should be considered. Generalizability should be explored.

Conclusion

Across the observed country-year data, electricity access was associated with stronger socio- economic and health outcomes. Basic access to drinking-water access was associated with lower infant mortality. These findings highlight important developmental factors contributing towards the GDP growth of a country.

References

Burke PJ, Stern DI, Bruns SB (2018), "The Impact of Electricity on Economic Development: A Macroeconomic Perspective". *International Review of Environmental and Resource Economics*, Vol. 12 No. 1 pp. 85–127, doi: https://doi.org/10.1561/101.00000101

Tong Zhang, Xunpeng Shi, Dayong Zhang, Junji Xiao,Socio-economic development and electricity access in developing economies: A long-run model averaging approach, Energy Policy,Volume 132, 2019, Pages 223-231, ISSN 0301-4215, https://doi.org/10.1016/j.enpol.2019.05.031.

Image sources https://unsplash.com/

Supplementary material

List of tables and figures

Figure 1

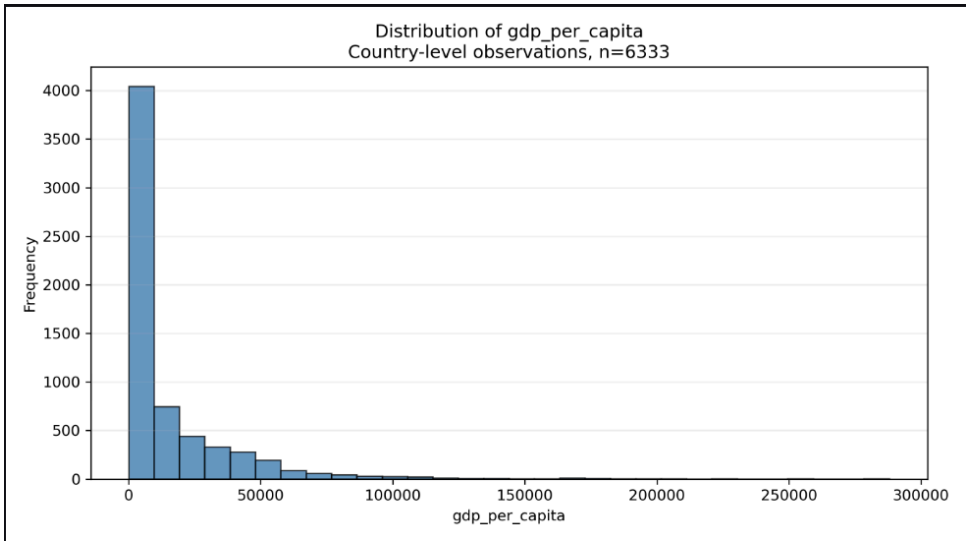


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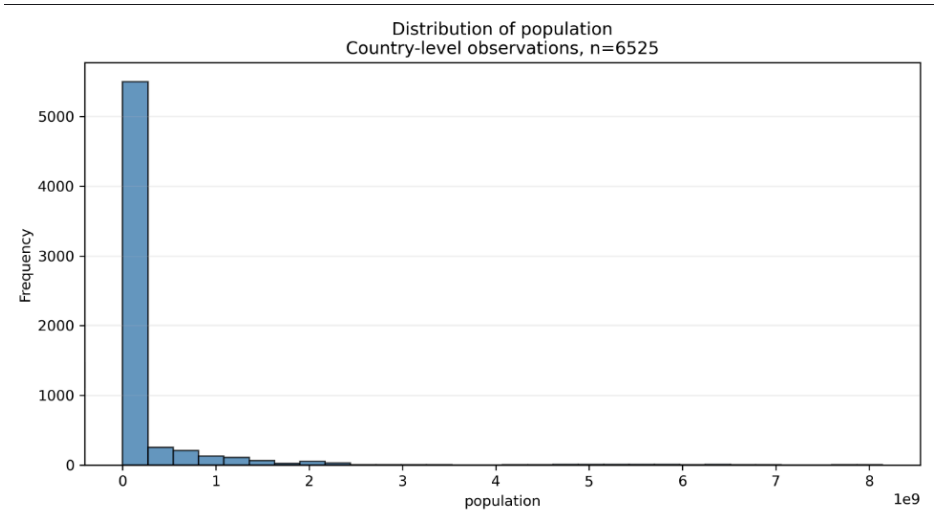


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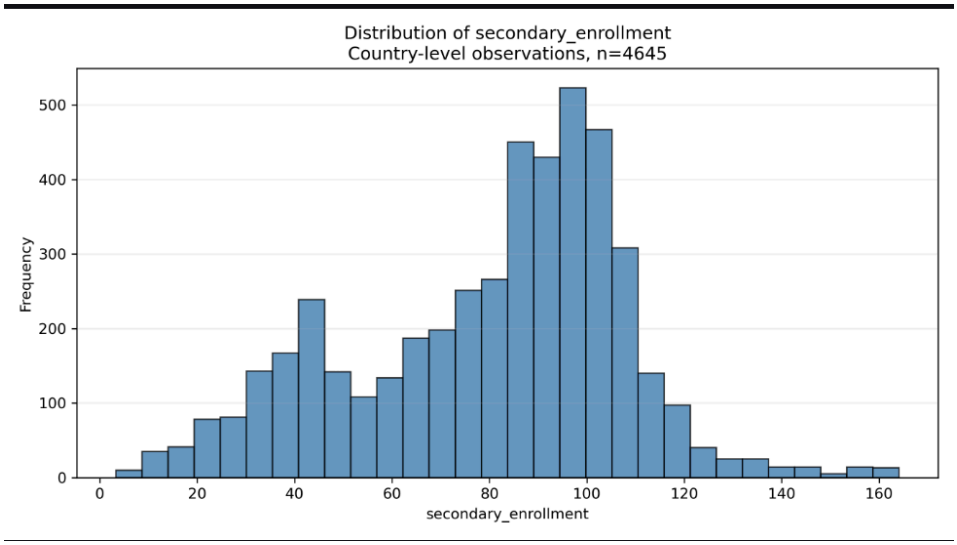


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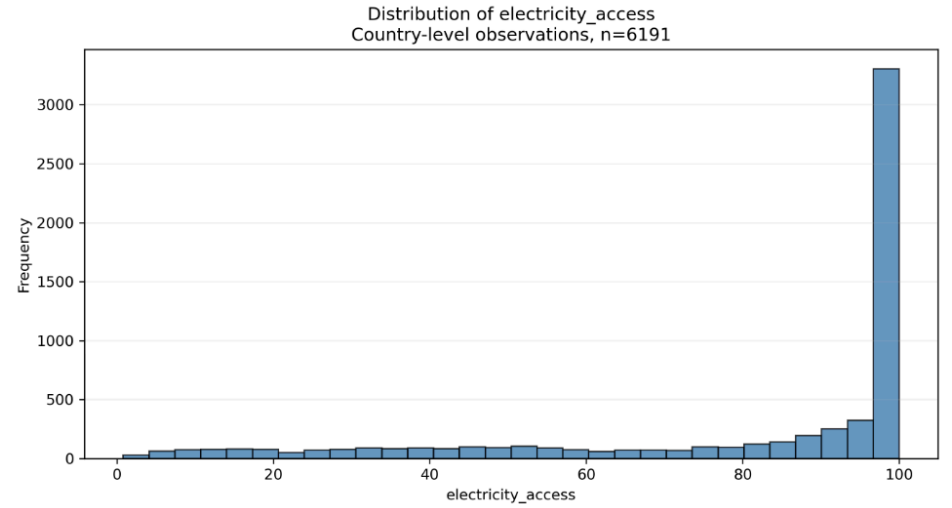


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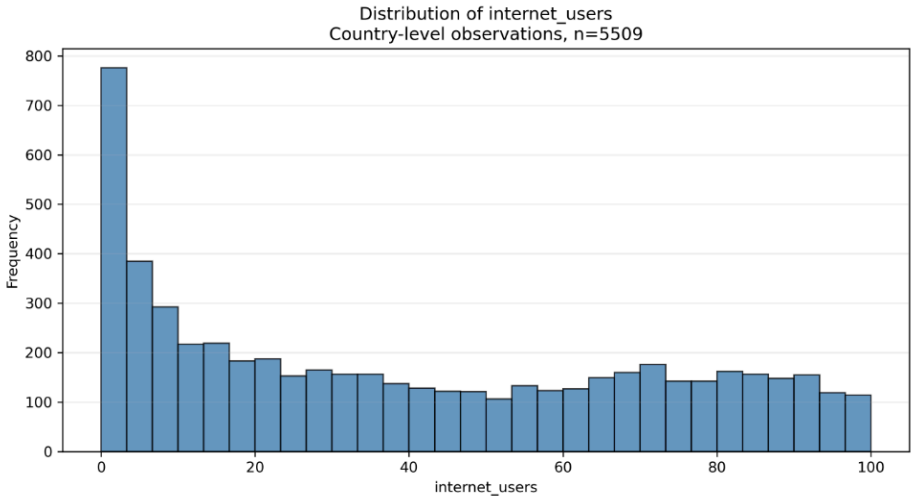


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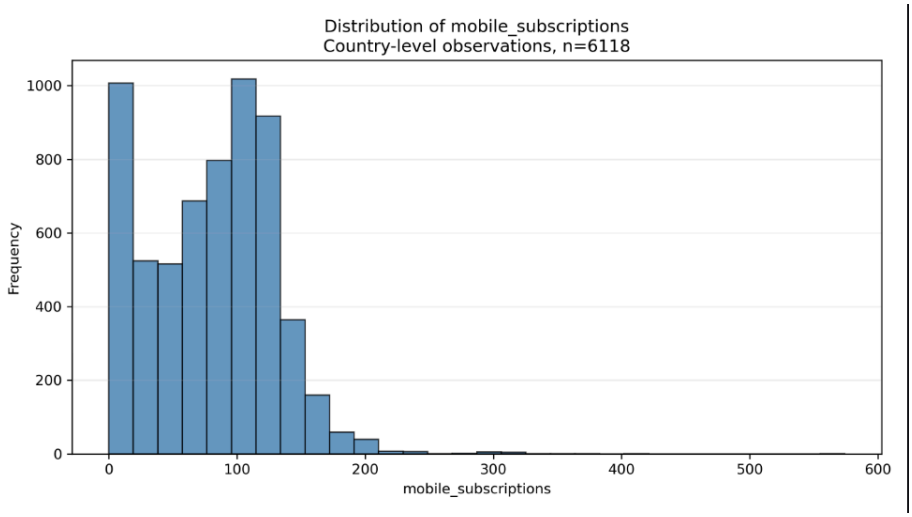


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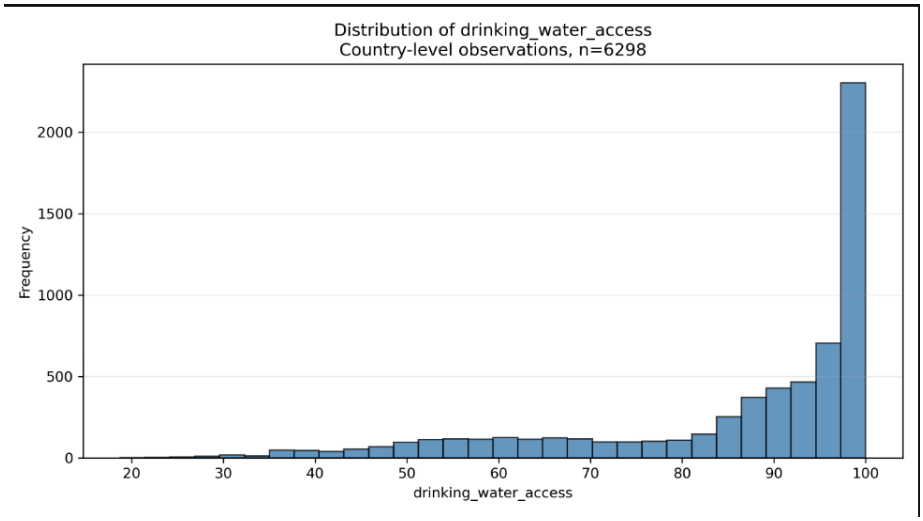


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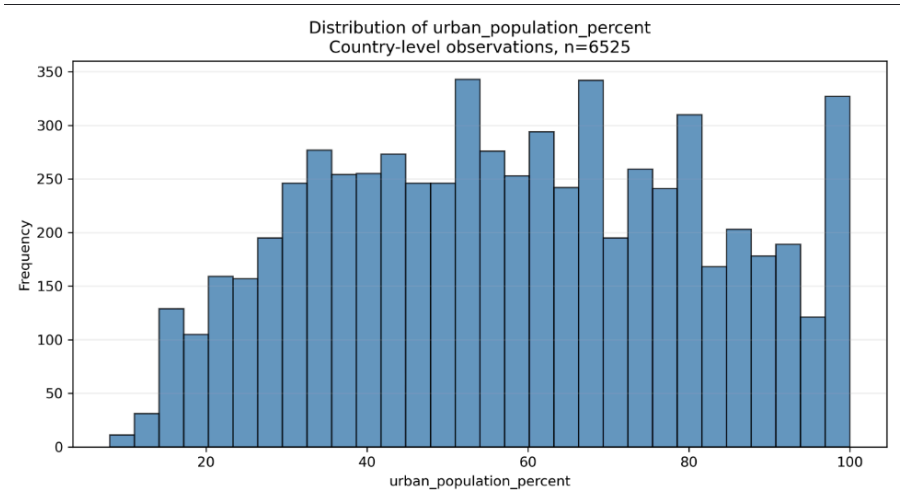


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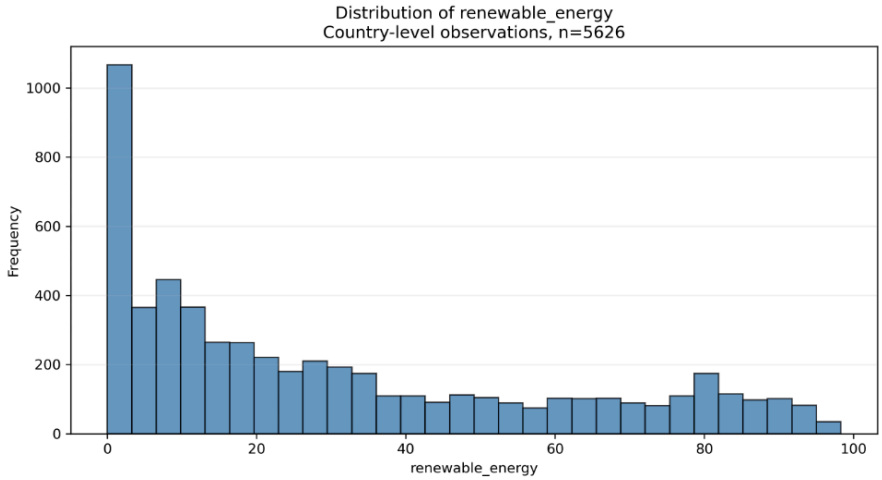


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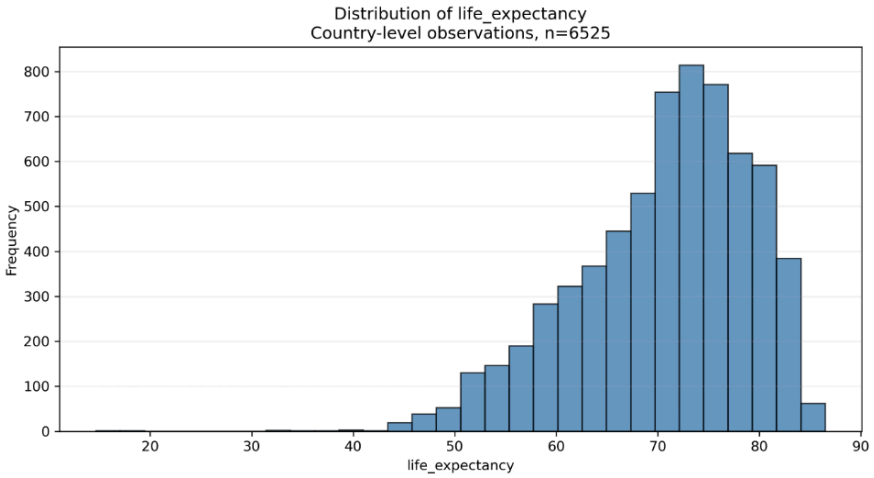


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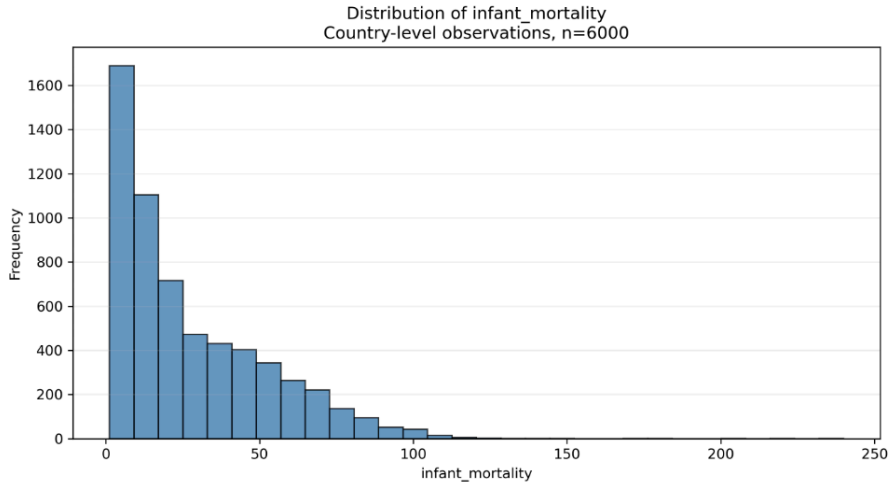


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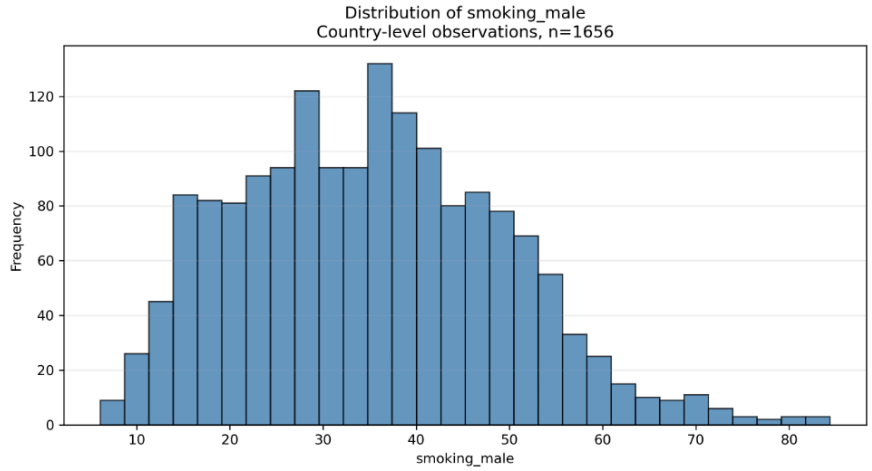


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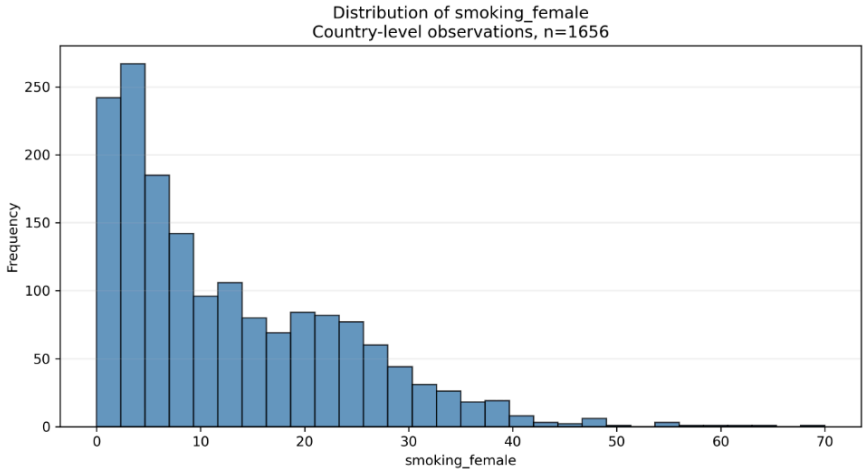


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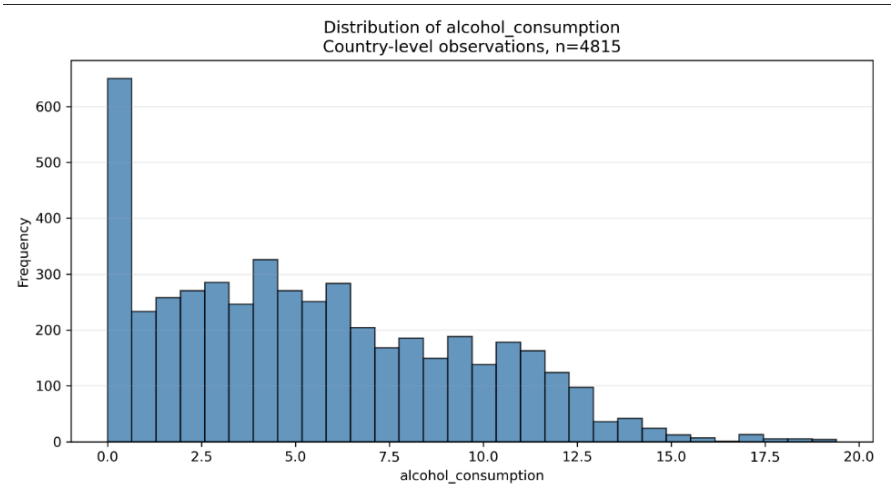


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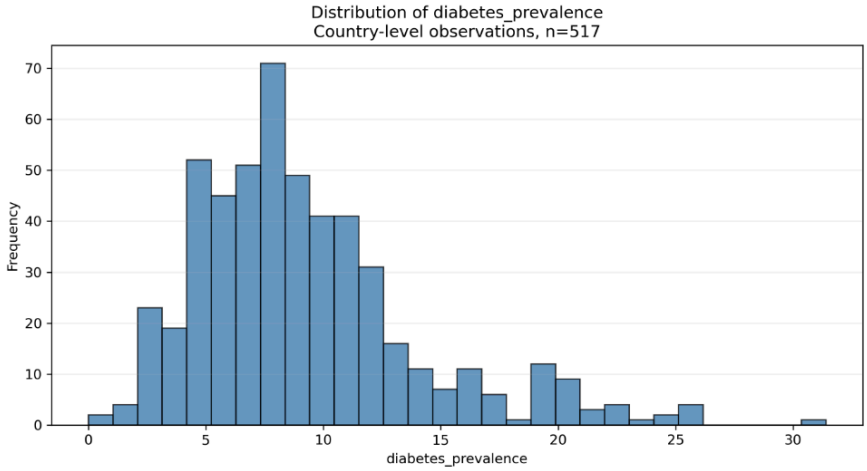


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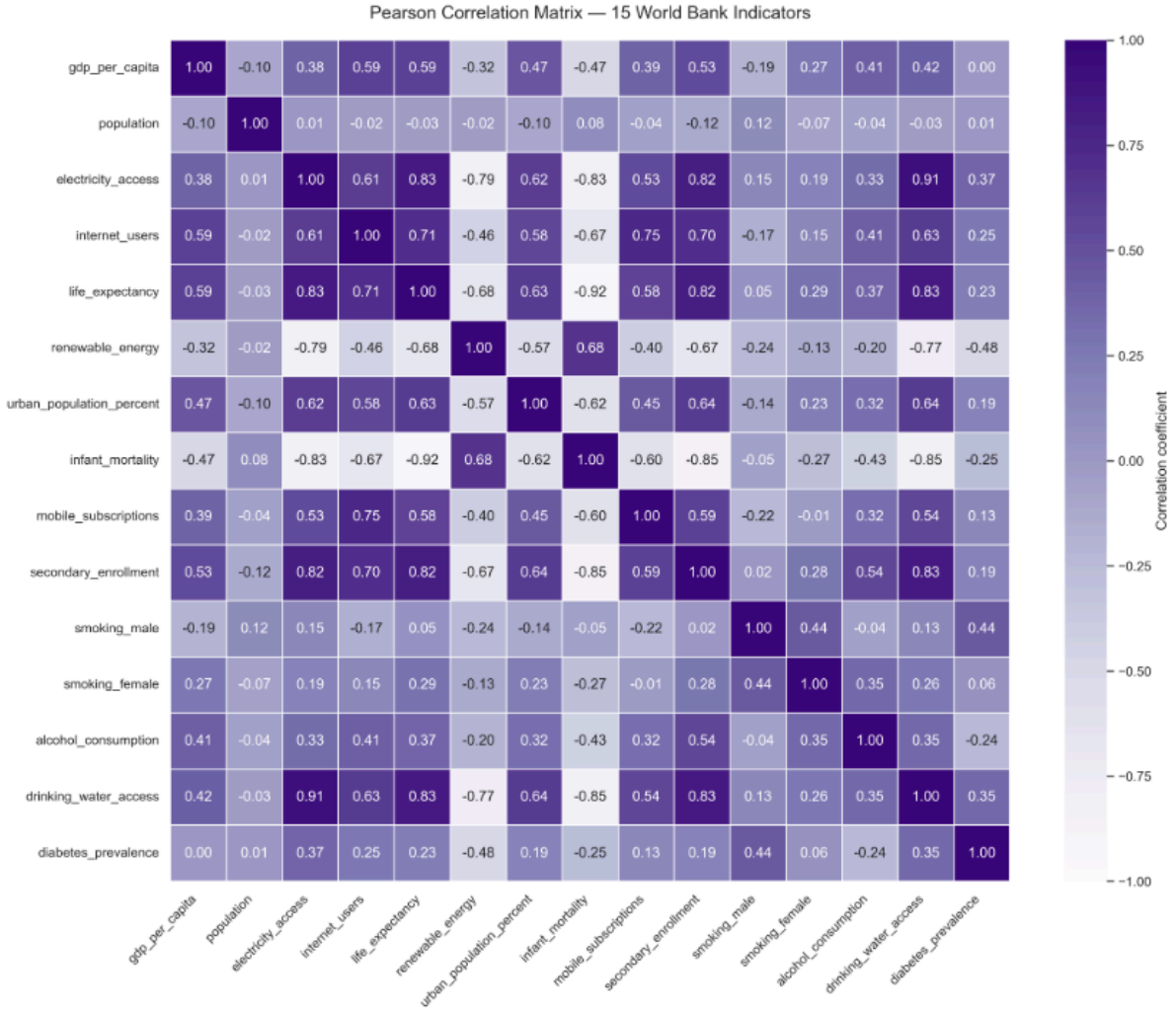


Figure 17


Spearman Correlation Matrix — 15 World Bank Indicators
gdp_per_capita 1.00 -0.27 0.85 0.80 0.87 -0.64 0.71 -0.86 0.66 0.81 -0.09 0.32 0.51 0.87 0.27
population -0.27 1.00 -0.27 -0.10 -0.19 0.27 -0.16 0.20 -0.08 -0.25 0.03 -0.09 -0.01 -0.32 -0.16
electricity_access 0.85 -0.27 1.00 0.77 0.87 -0.69 0.68 -0.87 0.60 0.82 0.08 0.27 0.42 0.88 0.26
internet_users 0.80 -0.10 0.77 1.00 0.76 -0.46 0.59 -0.77 0.81 0.74 -0.15 0.13 0.42 0.73 0.27
life_expectancy 0.87 -0.19 0.87 0.76 1.00 -0.58 0.66 -0.94 0.60 0.80 0.03 0.34 0.42 0.87 0.25
renewable_energy -0.64 0.27 -0.69 -0.46 -0.58 1.00 -0.57 0.57 -0.37 -0.54 -0.23 -0.08 -0.12 -0.66 -0.54
urban_population_percent 0.71 -0.16 0.68 0.59 0.66 -0.57 1.00 -0.69 0.46 0.64 -0.12 0.23 0.35 0.70 0.22
infant_mortality -0.86 0.20 -0.87 -0.77 -0.94 0.57 -0.69 1.00 -0.63 -0.85 -0.05 -0.39 -0.51 -0.88 -0.21
mobile_subscriptions 0.66 -0.08 0.60 0.81 0.60 -0.37 0.46 -0.63 1.00 0.63 -0.18 0.00 0.34 0.57 0.16
secondary_enrollment 0.81 -0.25 0.82 0.74 0.80 -0.54 0.64 -0.85 0.63 1.00 -0.01 0.37 0.59 0.82 0.16
smoking_male -0.09 0.03 0.08 -0.15 0.03 -0.23 -0.12 -0.05 -0.18 -0.01 1.00 0.44 -0.05 0.06 0.43
smoking_female 0.32 -0.09 0.27 0.13 0.34 -0.08 0.23 -0.39 0.00 0.37 0.44 1.00 0.39 0.37 -0.03
alcohol_consumption 0.51 -0.01 0.42 0.42 0.42 -0.12 0.35 -0.51 0.34 0.59 -0.05 0.39 1.00 0.44 -0.19
drinking_water_access 0.87 -0.32 0.88 0.73 0.87 -0.66 0.70 -0.88 0.57 0.82 0.06 0.37 0.44 1.00 0.30
diabetes_prevalence 0.27 -0.16 0.26 0.27 0.25 -0.54 0.22 -0.21 0.16 0.16 0.43 -0.03 -0.19 0.30 1.00
gdp_per_capita
population
electricity_access
internet_users
life_expectancy
renewable_energy
urban_population_percent
infant_mortality
mobile_subscriptions
secondary_enrollment
smoking_male
smoking_female
alcohol_consumption
drinking_water_access
diabetes_prevalence
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Correlation coefficient

Panel data plots

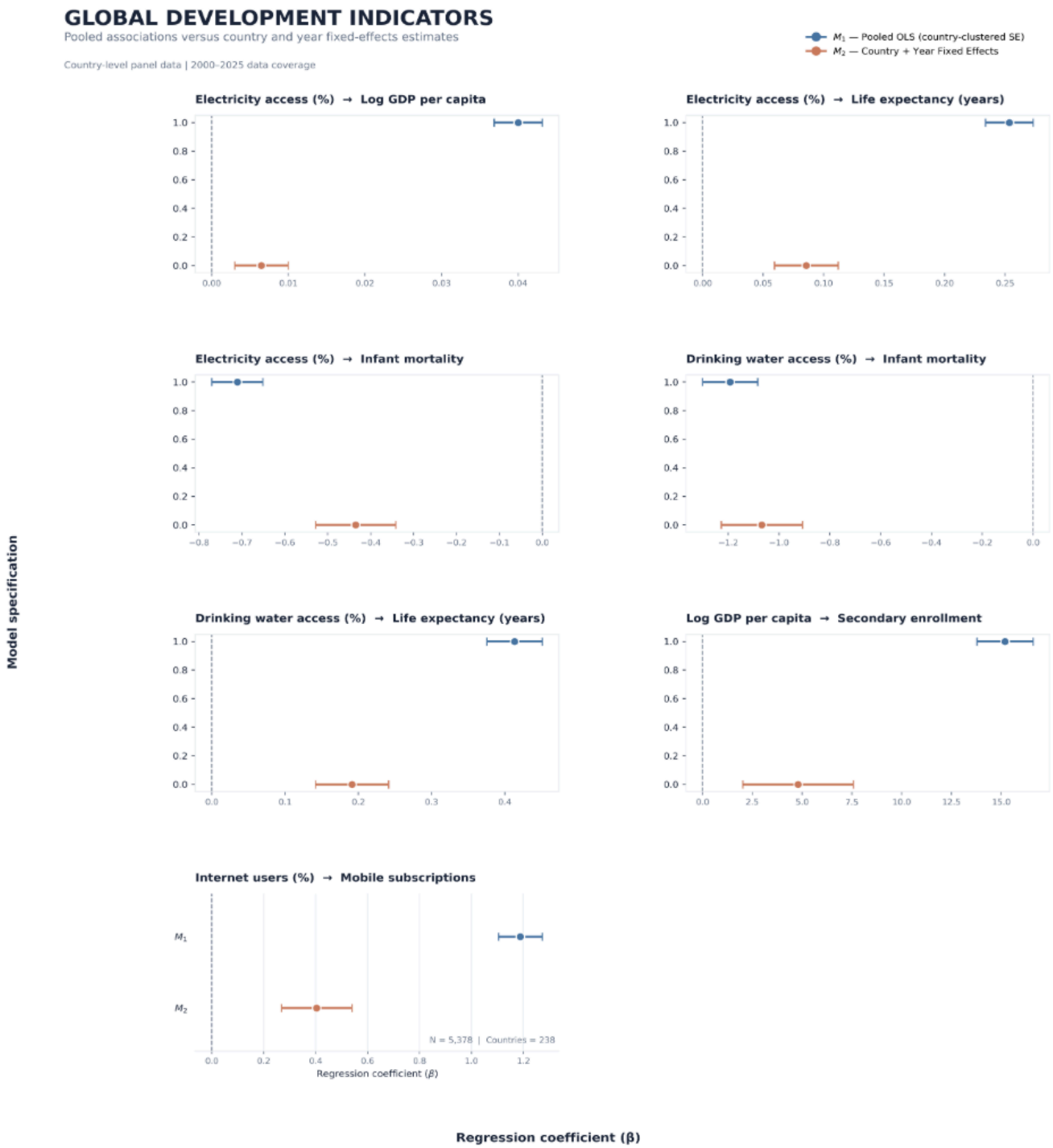


Table 1
Trends and development patterns for electricity access on a global level:

Dataset	Details
Number of rows	6,916
Number of columns	18
Name of entities	266
Country level observations	6,786
Observations(not indicating country level data)	130
Entities(not indicating country level data)	5
Period	2000 - 2023
Indicators used for the report	15

Table 2

Indicator	**Missing observations**	**Missing observations (%)**
1.GDP per capita , USD	**483**	**6.98**
2.Population	**291**	**4.21**
3.Secondary school enrollment gross (%)	**629**	**9.09**
4.Access to Electricity (% of population)	**1,323**	**19.13**
5.Use of Internet(% of population)	**291**	**4.21**
6.Mobile cellular subscriptions	**1,206**	**17.44**
7.Access to basic drinking water (% of population)	**291**	**4.21**
8.Urban population (% of total population)	**816**	**11.80**
9.Renewable	**754**	**10.90**

energy consumption		
10.Life expectancy at birth	**2,171**	**31.39**
11.Infant mortality per 1000 live births	**5,228**	**75.59**
12.Smoking prevalence in men	**5,228**	**75.59**
13.Smoking prevalence in women	**2,017**	**29.16**
14.Alcohol consumption per capita	**518**	**7.49**
15.Diabetes prevalence	**6,391**	**92.41**

Table 3

Development indicators selected for the purpose of crafting this report:

Development domain	Development indicator	World Bank Indicator (Code)
Economic	GDP per capita , USD	NY.GDP.PCAP.CD
Demographic	Population	SP.POP.TOTL
Education	Secondary school enrollment gross (%)	SE.SEC.ENRR
Infrastructure	Access to Electricity (% of population)	EG.ELC.ACCS.ZS
Infrastructure-Digital Connectivity	Use of Internet(% of population)	IT.NET.USER.ZS
Infrastructure-Digital Connectivity	Mobile cellular subscriptions	IT.CEL.SETS.P2
Infrastructure	Access to basic drinking water (% of population)	SH.H2O.BASW.ZS

Urban Development	Urban population (% of total population)	SP.URB.TOTL.IN.ZS
Energy consumption	Renewable energy consumption	EG.FEC.RNEW.ZS
Health	Life expectancy at birth	SP.DYN.LE00.IN
Health	Infant mortality per 1000 live births	SP.DYN.IMRT.IN
Health	Smoking prevalence in men	SH.PRV.SMOK.MA
Health	Smoking prevalence in women	SH.PRV.SMOK.FE
Health	Alcohol consumption per capita	SH.ALC.PCAP.LI
Health	Diabetes prevalence	SH.STA.DIAB.ZS

Table 4

Indicator	Observation after missingness	mean	median	Std deviation
1.GDP per capita , USD	6333	14,968.643	5,157.394	23,681.362
2.Population	6525	266,106,826.895	9,441,171.000	881,930,058.343
3.Secondary school enrollment gross (%)	4645	80.071	86.686	27.966
4.Access to Electricity (% of population)	6191	80.939	97.926	28.111
5.Use of Internet(% of population)	5509	39.619	33.895	31.641

6.Mobile cellular subscriptions	6118	79.515	84.800	49.738
7.Access to basic drinking water (% of population)	6298	86.151	87.873	17.275
8.Urban population (% of total population)	6525	51.986	57.959	23.018
9.Renewable energy consumption	5626	30.357	20.150	28.779
10.Life expectancy at birth	6525	70.648	72.062	8.559
11.Infant mortality per 1000 live births	6000	27.773	19.026	24.693

12.Smoking prevalence in men	1656	38.632	34.900	14.125
13.Smoking prevalence in women	1656	12.047	9.200	10.944
14.Alcohol consumption per capita	4815	5.387	4.883	3.976
15.Diabetes prevalence	517	**92.41**	93.774	4.816

Table 5

Indicator	**n**	**Pearson's r**	**Spearman's ρ**
Life expectancy × infant mortality	6000	-0.921	-0.937
Electricity × drinking water	6010	+0.909	0.884
Infant mortality × drinking water	5823	-0.846	-0.883
Infant mortality × electricity access	5711	-0.827	-0.872
Electricity × life expectancy	6051	+0.830	-0.872
Secondary enrollment × drinkingwater	5378	+0.829	0.813

Table 6

Y, X	n	Total countries	Pooled OLS Beta	Country and Year Fixed Effected
Electricity, log GDP per capita	6,051	256	0.0400	0.0065
Electricity, life expectancy	6,191	259	0.2535	0.0857
Electricity, infant mortality	5,711	239	-0.7108	-0.4351
Basic water, infant mortality	5,823	238	-1.1924	-1.0674
Basic water, life expectancy	6,298	258	0.4133	0.1914
Log GDP secondary enrollment	4,572	242	15.1889	4.7948
internetusers,mob sub	5,378	-	1.1883	0.4037